

▼ Data Cleaning

```
%pip install bertopic sentence-transformers
```

```
Requirement already satisfied: bertopic in /usr/local/lib/python3.12/dist-packages (0.17.3)
Requirement already satisfied: sentence-transformers in /usr/local/lib/python3.12/dist-packages (5.1.0)
Requirement already satisfied: hdbscan>=0.8.29 in /usr/local/lib/python3.12/dist-packages (from bertopic) (0.8.40)
Requirement already satisfied: umap-learn>=0.5.0 in /usr/local/lib/python3.12/dist-packages (from bertopic) (0.5.9.post2)
Requirement already satisfied: numpy>=1.20.0 in /usr/local/lib/python3.12/dist-packages (from bertopic) (2.0.2)
Requirement already satisfied: pandas>=1.1.5 in /usr/local/lib/python3.12/dist-packages (from bertopic) (2.2.2)
Requirement already satisfied: plotly>=4.7.0 in /usr/local/lib/python3.12/dist-packages (from bertopic) (5.24.1)
Requirement already satisfied: scikit-learn>=1.0 in /usr/local/lib/python3.12/dist-packages (from bertopic) (1.6.1)
Requirement already satisfied: tqdm>=4.41.1 in /usr/local/lib/python3.12/dist-packages (from bertopic) (4.67.1)
Requirement already satisfied: llvmlite>0.36.0 in /usr/local/lib/python3.12/dist-packages (from bertopic) (0.43.0)
Requirement already satisfied: transformers<5.0.0,>=4.41.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (4.41.0)
Requirement already satisfied: torch>=1.11.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (2.8.0+cu121)
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (1.16.1)
Requirement already satisfied: huggingface-hub>=0.20.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (0.20.0)
Requirement already satisfied: Pillow in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (11.3.0)
Requirement already satisfied: typing_extensions>=4.5.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (4.5.0)
Requirement already satisfied: joblib>=1.0 in /usr/local/lib/python3.12/dist-packages (from hdbscan>=0.8.29->bertopic) (1.5.2)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->sentence-transformers) (3.16.1)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->sentence-transformers) (2023.12.2)
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Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.1.5->bertopic) (2.9.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.1.5->bertopic) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.1.5->bertopic) (2025.2)
Requirement already satisfied: tenacity>=6.2.0 in /usr/local/lib/python3.12/dist-packages (from plotly>=4.7.0->bertopic) (8.5.0)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=1.0->bertopic) (3.5.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transformers) (75.1.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transformers) (1.13.3)
Requirement already satisfied: networkx in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transformers) (3.4.2)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transformers) (3.1.4)
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.6.77 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transformers) (12.6.77)
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.6.77 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transformers) (12.6.77)
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.6.80 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transformers) (12.6.80)
Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transformers) (9.10.2.21)
Requirement already satisfied: nvidia-cublas-cu12==12.6.4.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transformers) (12.6.4.1)
Requirement already satisfied: nvidia-cufft-cu12==11.3.0.4 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transformers) (11.3.0.4)
Requirement already satisfied: nvidia-curand-cu12==10.3.7.77 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transformers) (10.3.7.77)
Requirement already satisfied: nvidia-cusolver-cu12==11.7.1.2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transformers) (11.7.1.2)
Requirement already satisfied: nvidia-cusparse-cu12==12.5.4.2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-transformers) (12.5.4.2)
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Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers<5.0.0,>=4.41.0->bertopic) (2024.10.16)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers<5.0.0,>=4.41.0->bertopic) (0.20.1)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers<5.0.0,>=4.41.0->bertopic) (0.5.1)
Requirement already satisfied: numba>=0.51.2 in /usr/local/lib/python3.12/dist-packages (from umap-learn>=0.5.0->bertopic) (0.60.0)
Requirement already satisfied: pynndescent>=0.5 in /usr/local/lib/python3.12/dist-packages (from umap-learn>=0.5.0->bertopic) (0.5.10)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas>=1.1.5->bertopic) (1.17.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=1.11.0->sentence-transformers) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch>=1.11.0->sentence-transformers) (3.0.2)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests->huggingface-hub>=0.20.0->sentence-transformers) (3.4.1)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests->huggingface-hub>=0.20.0->sentence-transformers) (3.10.1)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests->huggingface-hub>=0.20.0->sentence-transformers) (2.3.0)
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from bertopic import BERTopic
from sentence_transformers import SentenceTransformer
```

```
df = pd.read_csv("/content/drive/MyDrive/Apex_Legends_Steam_Reviews.csv")
df.head()
```

	id	language	review	created	voted_up	votes_up	comment_count	steam_purchase	recieved_for_free	written_during_ea
0	157354180	english	very good	2024-02-01 20:07:07	True	0	0	False	False	
1	157354070	english	why are gay !!!	2024-02-01 20:05:35	False	0	0	False	False	
2	157353743	english	good game, play it	2024-02-01 19:59:42	True	0	0	False	False	
3	157352109	english	DO NOT PLAY THIS GAME ITS A DOG SHIT, EA CANT ...	2024-02-01 19:32:40	False	0	0	False	False	
4	157351840	english	nigger\n	2024-02-01 19:24:26	True	0	0	False	False	

```
#Cleaning Data
#remove special characters from reviews
df["review"] = df["review"].str.replace(r'[^a-zA-Z0-9\s]', '', regex=True)

# Fill NaN values with empty strings before applying sentiment analysis
if df["review"].isnull().any():
    df.fillna({"review":""}, inplace=True)
```

```
from datetime import datetime
```

```
df["created"] = pd.to_datetime(df["created"]).dt.normalize()
df.tail()
```

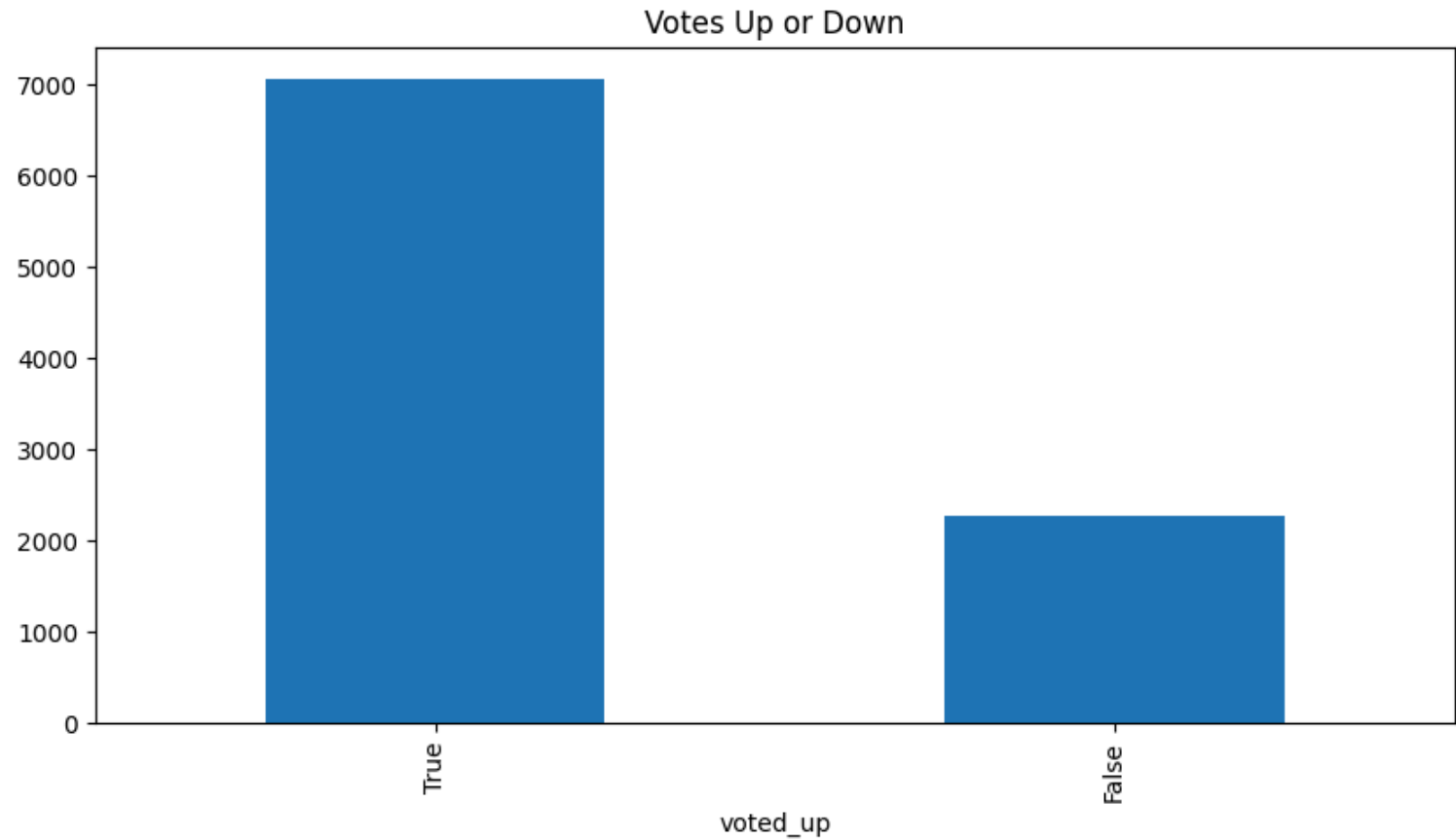
	id	language	review	created	voted_up	votes_up	comment_count	steam_purchase	recieved_for_free	written_during_ea
71310	129545870	english	stupid game	2022-12-28	False	1	0	False	False	
71311	129545713	english	its free	2022-12-28	True	0	0	False	False	
71312	129544748	english	good times	2022-12-28	True	0	0	False	False	
71313	129544646	english	They ruined this game a long time ago	2022-12-28	False	1	0	False	False	
71314	129543409	english	it gud for pew pew	2022-12-28	True	1	0	False	True	

```
#Split Data Into Seasons
seasons = {
    "Preseason": (datetime(2019, 2, 4),  datetime(2019, 3, 19)),
    "Season 1": (datetime(2019, 3, 19), datetime(2019, 6, 18)),
    "Season 2": (datetime(2019, 7, 2),  datetime(2019, 10, 1)),
    "Season 3": (datetime(2019, 10, 1), datetime(2020, 2, 4)),
    "Season 4": (datetime(2020, 2, 4),  datetime(2020, 5, 12)),
    "Season 5": (datetime(2020, 5, 12), datetime(2020, 8, 18)),
    "Season 6": (datetime(2020, 8, 18), datetime(2020, 11, 4)),
    "Season 7": (datetime(2020, 11, 4), datetime(2021, 2, 2)),
    "Season 8": (datetime(2021, 2, 2),  datetime(2021, 5, 4)),
    "Season 9": (datetime(2021, 5, 4),  datetime(2021, 8, 3)),
    "Season 10": (datetime(2021, 8, 3),  datetime(2021, 11, 2)),
    "Season 11": (datetime(2021, 11, 2), datetime(2022, 2, 8)),
    "Season 12": (datetime(2022, 2, 8),  datetime(2022, 5, 10)),
    "Season 13": (datetime(2022, 5, 10), datetime(2022, 8, 9)),
    "Season 14": (datetime(2022, 8, 9),  datetime(2022, 11, 1)),
    "Season 15": (datetime(2022, 11, 1), datetime(2023, 2, 14)),
    "Season 16": (datetime(2023, 2, 14), datetime(2023, 5, 9)),
    "Season 17": (datetime(2023, 5, 9),  datetime(2023, 8, 8)),
    "Season 18": (datetime(2023, 8, 8),  datetime(2023, 10, 31)),
    "Season 19": (datetime(2023, 10, 31), datetime(2024, 2, 13)),
}
```

```
"Season 20": (datetime(2024, 2, 13), datetime(2024, 5, 6)),
"Season 21": (datetime(2024, 5, 7), datetime(2024, 8, 6)),
"Season 22": (datetime(2024, 8, 6), datetime(2024, 11, 5)),
"Season 23": (datetime(2024, 11, 5), datetime(2025, 2, 11)),
"Season 24": (datetime(2025, 2, 11), datetime(2025, 5, 6)),
"Season 25": (datetime(2025, 5, 6), datetime(2025, 7, 5)),
}
```

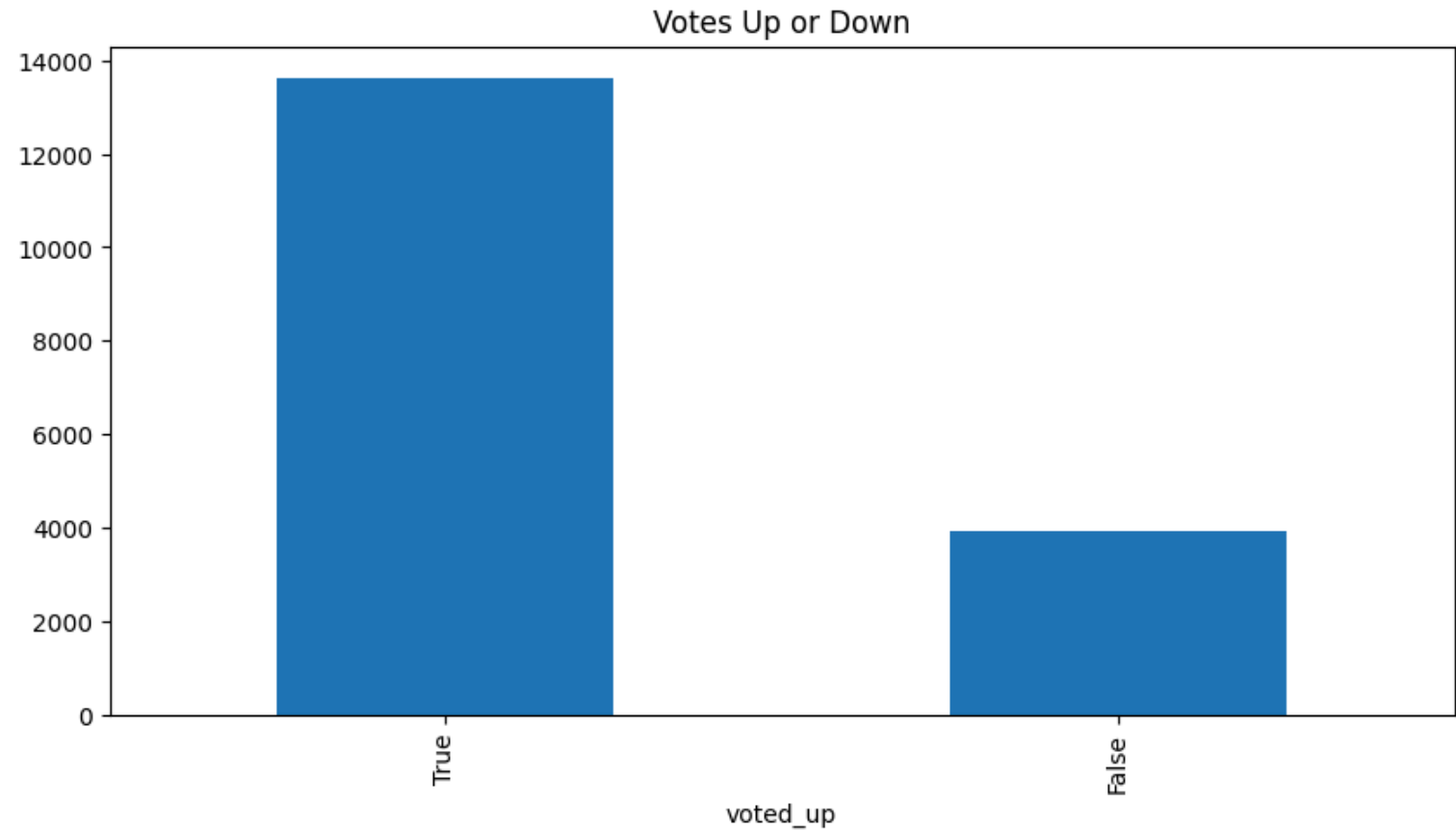
```
#Data Starts at Season 15
start, end = seasons["Season 15"]
s15_mask = (df['created'] >= start) & (df['created'] <= end)
df_s15 = df.loc[s15_mask]
df_s15["voted_up"].value_counts().plot(kind= "bar",
    title = "Votes Up or Down",
    figsize = (10,5))
```

<Axes: title={'center': 'Votes Up or Down'}, xlabel='voted_up'>



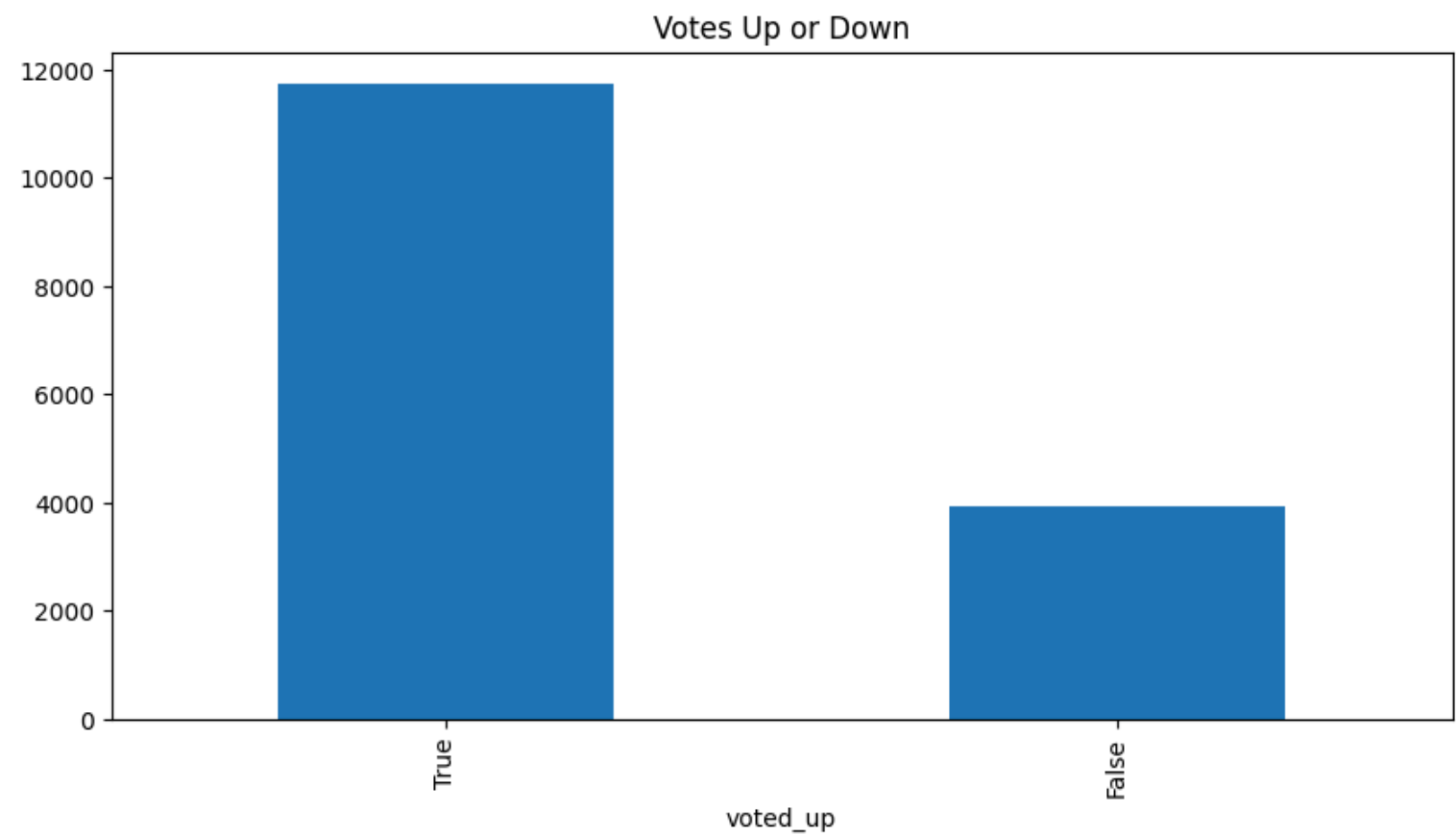
```
start, end = seasons["Season 16"]
s16_mask = (df['created'] >= start) & (df['created'] <= end)
df_s16 = df.loc[s16_mask]
df_s16.tail()
df_s16["voted_up"].value_counts().plot(kind= "bar",
    title = "Votes Up or Down",
    figsize = (10,5))
```

<Axes: title={'center': 'Votes Up or Down'}, xlabel='voted_up'>



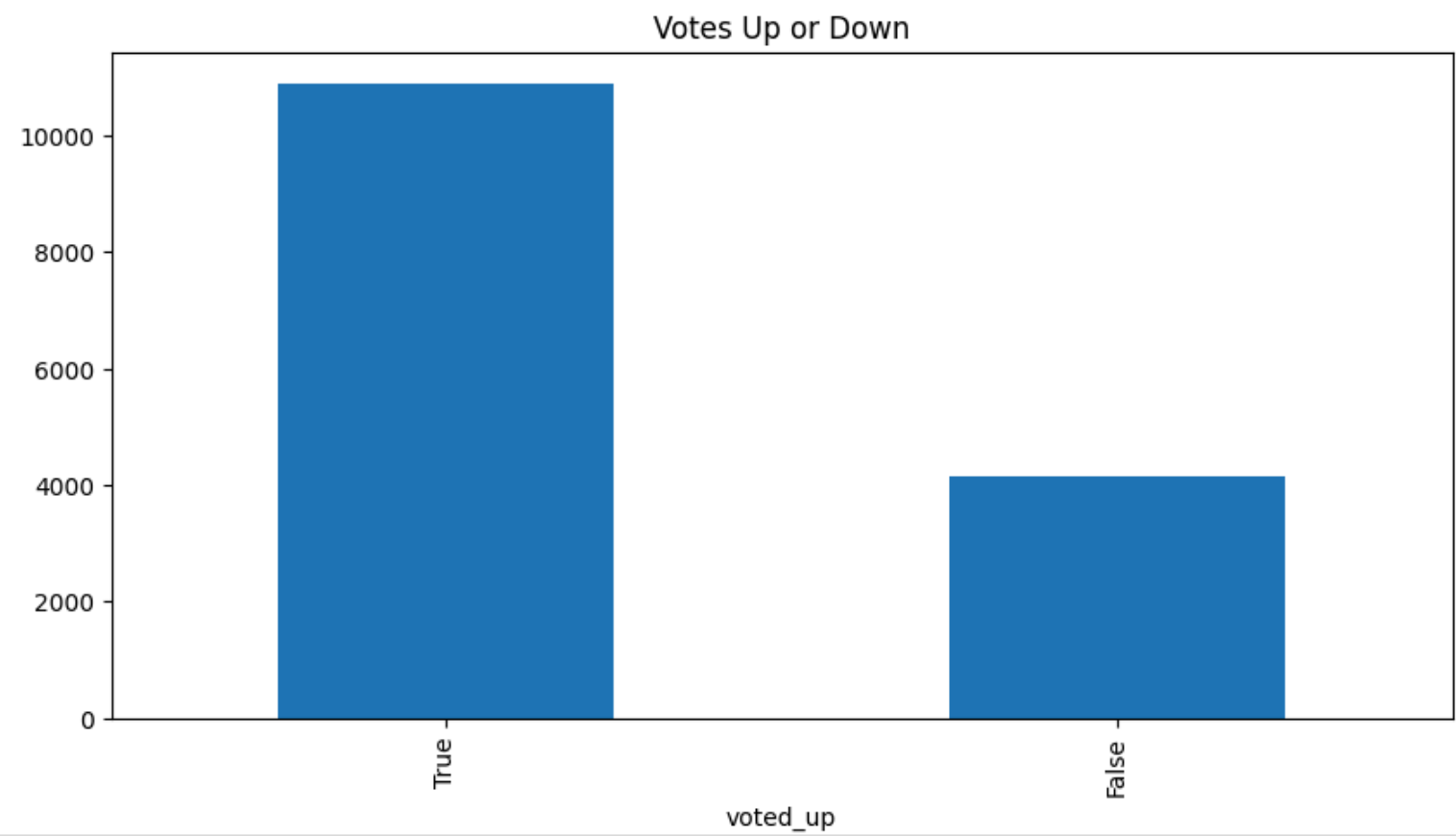
```
start, end = seasons["Season 17"]
s17_mask = (df['created'] >= start) & (df['created'] <= end)
df_s17 = df.loc[s17_mask]
df_s17.tail()
df_s17["voted_up"].value_counts().plot(kind= "bar",
    title = "Votes Up or Down",
    figsize = (10,5))
```

<Axes: title={'center': 'Votes Up or Down'}, xlabel='voted_up'>



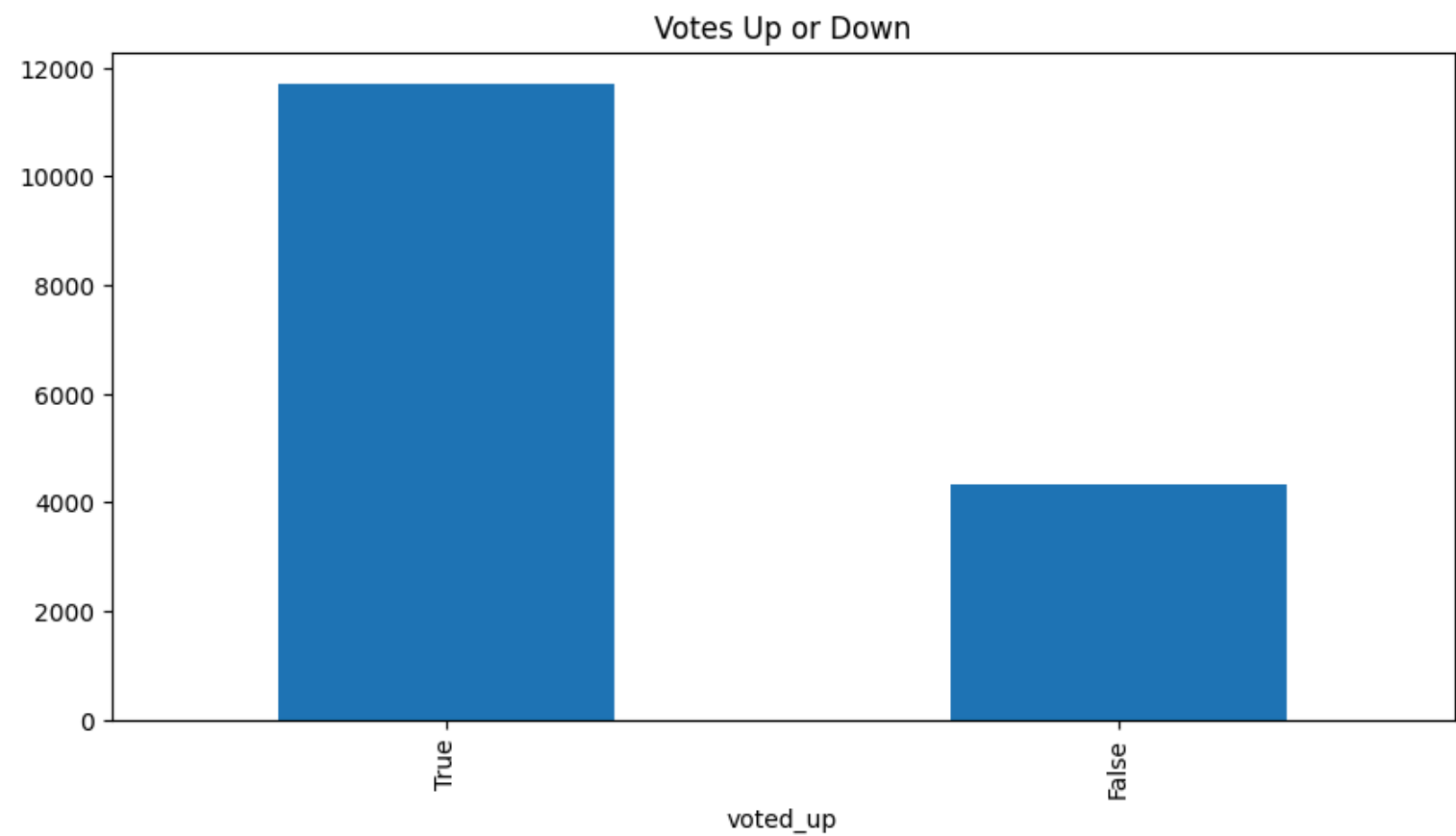
```
start, end = seasons["Season 18"]
s18_mask = (df['created'] >= start) & (df['created'] <= end)
df_s18 = df.loc[s18_mask]
df_s18.tail()
df_s18["voted_up"].value_counts().plot(kind= "bar",
    title = "Votes Up or Down",
    figsize = (10,5))
```

<Axes: title={'center': 'Votes Up or Down'}, xlabel='voted_up'>



```
start, end = seasons["Season 19"]
s19_mask = (df['created'] >= start) & (df['created'] <= end)
df_s19 = df.loc[s19_mask]
df_s19.head()
#Data Ends at Season 19
df_s19["voted_up"].value_counts().plot(kind= "bar",
    title = "Votes Up or Down",
    figsize = (10,5))
```

<Axes: title={'center': 'Votes Up or Down'}, xlabel='voted_up'>



Clustering

df_s15.head()

	id	language	review	created	voted_up	votes_up	comment_count	steam_purchase	recieved_for_free	written_du
61998	132862276	english	g	2023-02-14	True	0	0	False	False	
61999	132862220	english	Can be highly competitive or casual with multi...	2023-02-14	True	0	0	False	False	
62000	132862134	english	Awesome game have a blast playing w friends	2023-02-14	True	0	0	False	True	
62001	132862133	english	but takes to much space\n	2023-02-14	True	0	0	False	False	
62002	132861925	english	trash lol	2023-02-14	False	0	0	False	False	

embedding_model = SentenceTransformer("all-MiniLM-L6-v2")

modules.json: 100% 349/349 [00:00<00:00, 16.0kB/s]
config_sentence_transformers.json: 100% 116/116 [00:00<00:00, 6.65kB/s]
README.md: 10.5k/? [00:00<00:00, 799kB/s]
sentence_bert_config.json: 100% 53.0/53.0 [00:00<00:00, 4.64kB/s]
config.json: 100% 612/612 [00:00<00:00, 46.2kB/s]
model.safetensors: 100% 90.9M/90.9M [00:01<00:00, 83.9MB/s]
tokenizer_config.json: 100% 350/350 [00:00<00:00, 31.0kB/s]
vocab.txt: 232k/? [00:00<00:00, 2.97MB/s]
tokenizer.json: 466k/? [00:00<00:00, 6.38MB/s]
special_tokens_map.json: 100% 112/112 [00:00<00:00, 8.84kB/s]
config.json: 100% 190/190 [00:00<00:00, 15.0kB/s]

```
topic_model = BERTopic(  
    embedding_model=embedding_model,  
    verbose=True,  
    min_topic_size=50    # merge small clusters  
)
```

reviews = df["review"].tolist()

```
topics, probs = topic_model.fit_transform(reviews)
```

```
# Store topic assignments
df["topic"] = topics
```

2025-09-16 00:42:31,165 - BERTopic - Embedding - Transforming documents to embeddings.
Batches: 100% 2229/2229 [17:49<00:00, 22.69it/s]
2025-09-16 01:00:21,657 - BERTopic - Embedding - Completed ✓
2025-09-16 01:00:21,659 - BERTopic - Dimensionality - Fitting the dimensionality reduction algorithm
2025-09-16 01:04:56,725 - BERTopic - Dimensionality - Completed ✓
2025-09-16 01:04:56,729 - BERTopic - Cluster - Start clustering the reduced embeddings
2025-09-16 01:05:16,954 - BERTopic - Cluster - Completed ✓
2025-09-16 01:05:16,992 - BERTopic - Representation - Fine-tuning topics using representation models.
2025-09-16 01:05:19,201 - BERTopic - Representation - Completed ✓

▼ Sentiment Analysis

```
from transformers import pipeline
```

```
sentiment_model = pipeline(
    "sentiment-analysis",
    model="cardiffnlp/twitter-roberta-base-sentiment"
)
```

config.json: 100% 747/747 [00:00<00:00, 20.9kB/s]
pytorch_model.bin: 100% 499M/499M [00:14<00:00, 59.8MB/s]
model.safetensors: 100% 499M/499M [00:18<00:00, 17.0MB/s]
vocab.json: 899k/? [00:00<00:00, 6.37MB/s]
merges.txt: 456k/? [00:00<00:00, 6.14MB/s]
special_tokens_map.json: 100% 150/150 [00:00<00:00, 1.87kB/s]
Device set to use cpu

```
#Get Sentiments for Batches
```

```
sentiments = []
batch_size = 32
for start in range(0, len(reviews), batch_size):
    chunk = reviews[start:start+batch_size]
    results = sentiment_model(chunk, truncation=True, max_length=512)
    sentiments.extend([r["label"] for r in results])

df["sentiment"] = sentiments
```

```
#Sentiment Summary per Topic
```

```
timestamps = df["created"].tolist()
```

```
sent_summary = (
    df.groupby("topic")["sentiment"]
    .value_counts(normalize=True)
    .unstack(fill_value=0)
    .sort_index()
)
```

sentiment	LABEL_0	LABEL_1	LABEL_2
topic			
-1	0.339837	0.377846	0.282317
0	0.000000	1.000000	0.000000
1	0.115356	0.076399	0.808245
2	0.000552	0.007174	0.992274
3	0.000000	0.946040	0.053960

```
topics_over_time = topic_model.topics_over_time(
    reviews, timestamps, nr_bins=20
)
```

20it [00:34, 1.71s/it]

```
topic_model.visualize_topics_over_time(topics_over_time)
```